

Riley Beenders

R&D, Electromechanical and Automation Engineer

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PROFILE

I'm an electromechanical engineer and R&D leader specializing in product development, manufacturing and industrial automation. I bring together mechanical design, electronics, controls, software, and manufacturing to develop reliable, practical products and systems built for real-world use. That same curiosity extends beyond my professional work - I love traveling the world, rebuilding cars, replicating movie props and finding creative ways to automate the repetitive parts of my everyday life. RileyBeenders.com is another extension of my curiosity - a platform I built to house my experience, skills, and projects in one place, create focused applications for ambitious opportunities, and demonstrate the full range of what I can offer.

EXPERIENCE

Lead, Research & Development Engineer

January 2025 - Present

Proteor / Proteor Printing Solutions | Coplay, PA

Proteor acquired Filament Innovations in January, 2025 and my role evolved from CTO to Lead R&D Engineer. I continue to carry out my former CTO responsibilities, now with expanded oversight of the East Coast engineering team and involvement across Proteor's global product portfolio.

- Led the design and development of the ICARUS-Lite, a compact 3D printer purpose-built for digital prosthetic and orthotic production. Primarily responsible for its mechanical, electrical, and software design, including custom AC heaters and dual-voltage 120/240 VAC operation.
- Led the ICARUS-Lite from product development through commercialization, contributing to increased revenue and the addition of approximately five new roles.
- Developed an automated LabVIEW system that collects nine data points every 15 ms from multiple laser sensors over Modbus and charts tolerance data for internal quality control and end-customer use.
- Oversight of Microsoft's Business Central software implementation for all of my division's items and products. The Business Central software syncs with all of our other Proteor locations around the world to make sales and managing inventory easier.
- Lead the implementation of ProteorPrint OS, a network-accessible remote operation and diagnostics interface with live video, AI-assisted alerts, material, waste, and outcome metrics. This software was developed in collaboration with SimplyPrint.io
- Performed and documented risk analysis covering safety, operational, and software risks for ICARUS and ICARUS-Lite products along with the ProteorPrint cloud platform, implementing mitigations applicable to EU and U.S. standards.
- Collaborated with another Additive Manufacturer to develop a reliable, cost effective hotend that meets our material's specific needs.
- Work alongside production, sales, and customer service teams to support the product line. Responsibilities include hardware, electronics, motion-control systems, software, troubleshooting and after-sales technical support.
- Converted the Filament Innovation POSEIDON and ICARUS platforms from servo-driven ballscrews to a standardized linear-motor architecture, achieving a 1-micron precision per 1 m/s while moving a 30 kg payload. The system maxed out at a speed of 4 m/s
- Continued development of our internal IT infrastructure and standardized software and internal processes within our division of Proteor.

Chief Technology Officer

January 2023 - January 2025

Filament Innovations | Whitehall, PA

Acquired by Proteor and is recognized globally as 'Proteor Print'.

- Oversaw the product-development pipeline, internal IT infrastructure, and technology strategy while retaining hands-on responsibility for mechanical, electrical, firmware, and software systems.
- Led the design and launch of the Gen3 POSEIDON, combining a 1 m³ build volume, filament-and-pellet dual extrusion, braked NEMA 34 Z drives, industrial servo XY motion, and a custom gantry.
- Designed and hosted a redundant Kubernetes cluster running a self-developed ERP/CRM for orders, inventory, tasks, projects, and customer interactions, with dynamic BOM generation linked to purchasing.
- Researched high-speed linear motors that moved a 30 kg payload at 3 m/s with 5-micron accuracy and conducted early 6-axis robotic non-planar printing research.

Lead Electro-Mechanical Engineer

June 2022 - January 2023

Filament Innovations | Whitehall, PA

- Improved thermal and acoustic performance through enclosure redesign and component placement, including custom tooling with local fabricators for large-radius aluminum-panel bends.
- Integrated a multi-board CAN-bus control architecture that simplified wiring and debugging while supporting larger, modular machines.
- Developed independently testable electronics enclosures that could be installed in minutes and coordinated custom harnesses, panel designs, and hardware improvements with vendors.

Industrial Engineer

April 2021 - June 2022

Filament Innovations | Whitehall, PA

- Designed and released Kratos, a machine engineered to print high-strength above-knee and below-knee prosthetic sockets.
- Developed next-generation ICARUS and POSEIDON models with higher-performance components and expanded motion safety through NEMA 34 motors, planetary gearboxes, and power-failure brakes.
- Designed THE KRAKEN, a 1 x 2 x 1.5 m hybrid pellet-and-filament gantry printer for a client producing the world's largest 3D-printed statue, and managed CAD, BOM, sourcing, fabrication, assembly, firmware tuning, support, and training.

Engineering Technician

May 2020 - April 2021

Filament Innovations | Whitehall, PA

- Diagnosed mechanical and electrical problems during assembly and operation, improving early product reliability.
- Tuned, calibrated, and optimized print parameters while documenting design limitations and opportunities for product improvement.

Owner & Engineering Consultant

2015 - Present

RJB.Engineering | Pennsylvania

- Deliver project-based engineering support for automotive clients and operate a small online store for 3D-printing heat inserts and other printer accessories.
- Produced 3D-printed plastic face shields during the COVID-19 pandemic.

Ski Lift Operator

November 2019 - January 2021

Blue Mountain Ski Resort | Palmerton, PA

- Operated lifts and assisted thousands of guests with ticketing, weather, and slope information.
- Performed motor and control safety checks and supported the maintenance team during lift breakdowns.

Computer / Network Technician

June 2016 - August 2019

Northampton Area School District | Northampton, PA

- Built network infrastructure and deployed more than 750 wireless access points across six school buildings.
- Accelerated deployment of more than 2,500 Chromebooks by developing an Arduino-based tool to automate setup-data entry.
- Coordinated final device distribution with school administrators.

EDUCATION

Penn State University

BS, Electro-Mechanical Engineering

December 2024

CERTIFICATIONS

MATLAB OnRamp

MathWorks | Issued July 2026

[Show credential](#)

Simulink OnRamp

MathWorks | Issued July 2026

[Show credential](#)

Simscape OnRamp

MathWorks | Issued July 2026

[Show credential](#)

Machine Learning Onramp

MathWorks | Issued July 2026

[Show credential](#)

Lean Six Sigma Yellow Belt

Educate 360 | Issued July 2025

[Show credential](#)

Lean Six Sigma White Belt

Educate 360 | Issued June 2025

[Show credential](#)

Certificate in Engineering Design and Tools

Penn State University | Issued Spring 2023

Mechanical Engineering Certificate

Lafayette College | Issued Spring 2019

SKILLS

Engineering

Design for Manufacturing, Electrical Layout and Schematic Design, Engineering Documentation and BOM Control, FDM, FGF, MJF, and SLS Design, Fixture and Tooling Design, Product Development and Commercialization, Product Safety and Regulatory Compliance, Production Workflow Optimization, R&D Prototyping & Validation, Technical Writing and Documentation, Vendor Communication and Cost Negotiation, Worldwide Project Collaboration

Design & Software

Autodesk Inventor, CATIA, Fixture and Tooling Design, Fusion 360, GitHub, LabVIEW, LM Studio, Macros, MATLAB, oMLX, Python, Rapid Prototyping, Siemens NX, SolidWorks, Visual Studio Code

Controls & Automation

Allen Bradley PLCs, Data Logging and Process Analysis, Git Version Control, Workflow Automation, Windows Applications

Infrastructure & Systems

Domain-Based Application Access, DNS and Subdomain Configuration, Docker and Containerized Applications, Proxmox Virtualization, Reverse Proxy and Firewall Routing, Self-Hosted Infrastructure, Server Network and VLAN Configuration, Virtual Machine and Container Management, Windows/Linux Systems Administration